

Yaseen Basha Thippaluri

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SUMMARY

Final-year AIML student who builds AI-assisted tools end to end, from data pipeline to live demo. Most of what I build starts from a problem I actually hit: a repo analyzer for unfamiliar 2,000-file codebases, an incident-memory tool because on-call knowledge walks out the door. Our RL agent placed 12th of 31,000+ registrations at the OpenEnv hackathon (Meta, Hugging Face, PyTorch). Three merged PRs into open-source AI tooling (TRL, LiteLLM).

PROJECTS

MemOps: Incident Memory for On-Call Engineers

Solo project

Python, FastAPI, Cognee knowledge graph, React, D3.js, Groq (Llama-3.3-70B)

[Github](#) · [Live Demo](#)

- Prototyped an AI-assisted incident workflow: resolved incidents feed a knowledge graph so the next on-call engineer doesn't start from zero. 17 incidents across 10 services, shown in a live D3 dashboard.
- Paste an alert to get past incidents ranked by real embedding similarity from Cognee's vector store, plus a synthesized fix. Approving it re-indexes the graph and shows a true before and after, never an invented diff.
- Cut an $O(n)$ LLM call fan-out by consolidating ingestion into one shared dataset, fixed file-handle leaks in long-running instances, and kept the LLM provider a swappable config value.

Enterprise IT Triage Agent: OpenEnv RL Environment

12th of 31,000+, Top 100 finalist

PyTorch, TRL/GRPO, Qwen2.5-3B, OpenEnv, Hugging Face

[Live Demo](#) · [Trained Model](#)

- Two-person team; I owned reward design and GRPO training. Six independent reward components (format, resolution, citation grounding, calibration, parsimony, repetition) mean gaming one costs on the others.
- Shipped the parsimony reward with an inverted branch; the model found it within four hours, padding answers with filler to score as if complete. Rebuilding that reward made the rest of the run work.
- Ran 200 GRPO steps on an A100, tracking six signals in Weights & Biases and tuning reward weights between runs. Calibration 0.53 to 0.98, parsimony 0.25 to 0.94.

Intelligent GitHub Repository Analyzer

Top 3 of 108 teams, PythonGuru × Supervity AI Hackathon

Python, Streamlit, LangChain, rank-bm25, NLTK, GitPython, Groq / Gemini / HF Inference APIs

[Github](#)

- Retrieval-augmented Q&A over codebases you've never seen. BM25 ranking picks the relevant context before it reaches the LLM, keeping responses under 5 seconds across the 50+ repositories I tested.
- Multi-LLM layer swappable across Groq, Gemini and Hugging Face Inference.

OPEN SOURCE

Hugging Face TRL (LLM training library): 2 merged PRs

[PR #6648](#) · [PR #6654](#)

- Fixed GRPOTrainer's entropy bonus inheriting the policy loss normalization, which silently changed regularization strength across loss types. Found while training the triage agent.
- Fixed the LuSPO loss not applying its completion mask elementwise before aggregation, which reintroduced the length bias it exists to remove.

LiteLLM (open-source LLM gateway, 59k+ stars): 1 merged PR

[PR #43197](#)

- Fixed Gemini models rejecting the seed parameter, which errored out or, with drop_params on, was silently discarded. Added tests for both modes and documented before and after proof against the live Gemini API.

SKILLS

AI & LLM: LLMs, RAG, agents, vector search, knowledge graphs, LangChain, PyTorch, RL fine-tuning (TRL/GRPO)

Languages: Python, JavaScript, Java, SQL, C

Backend: FastAPI, Node.js, REST APIs, PostgreSQL, MongoDB

Frontend: React, Tailwind CSS, D3.js, Streamlit

Tools & Platforms: Git, GitHub, Docker, Linux, Vercel, Hugging Face Spaces

AI-assisted development: Claude Code and GitHub Copilot, daily, including full-stack delivery of MemOps

EDUCATION & CERTIFICATIONS

Vignan's Lara Institute of Technology

Guntur, India

B.Tech, Artificial Intelligence and Machine Learning, CGPA 7.51

2022 – 2026

CERTIFICATIONS

AI Engineer Path (Scrimba) · Data Structures and Algorithms (Apna College)